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Subject Code: MPC203T

Roll No:

MPHARM**(SEM. II) THEORY EXAMINATION 2025-26****COMPUTER AIDED DRUG DESIGN****TIME: 3 HRS****M.MARKS: 75****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt *all* questions in brief.****10x2 = 20**

- (a) Define drug design.
- (b) What is SAR?
- (c) Define sigma constant.
- (d) What is steric parameter?
- (e) Define docking score.
- (f) What is bioactive conformation?
- (g) Define pharmacokinetics.
- (h) What is ligand-based design?
- (i) Define contour maps.
- (j) What is virtual library?

SECTION B**2. Attempt any *two* of the following:****2 x 10 = 20**

- (a) Explain Hansch Analysis in detail. Derive the Hansch equation and discuss its advantages and disadvantages over the Free Wilson model.
- (b) Detailed study of Molecular Docking. Explain the types of docking and the various drug-receptor interactions involved (Hydrogen bonding, Van der Waals, etc.).
- (c) Discuss the development of 2D-QSAR and 3D-QSAR models. Explain contour maps and statistical validation methods.

SECTION C**3. Attempt any five part of the following:****7 x 5 = 35**

- (a) What is Fragment-Based Drug Design (FBDD)? Compare it with High-Throughput Screening (HTS).
- (b) Describe the steps involved in Pharmacophore Modeling and conformational searching.
- (c) Discuss 3D-QSAR approaches. Explain the concept of CoMFA (Comparative Molecular Field Analysis).
- (d) Describe Pharmacophore Mapping. How are pharmacophore features identified?
- (e) Discuss In-silico Virtual Screening protocols. Differentiate between Structure-based and Ligand-based virtual screening.
- (f) Discuss ADMET properties and their importance in reducing drug failure during clinical trials.
- (g) Explain the mechanism of action and CADD studies of agents acting on DHFR or HIV Protease